NETWORKING

QUIC
Visual Guide

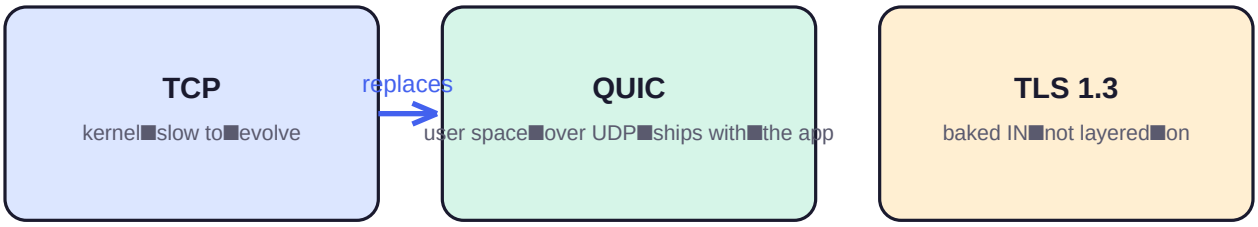
The UDP transport that grew up —
fast handshakes, migration, and HTTP/3.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

A Transport Over UDP

Reliability rebuilt in user space — deployable without kernel updates.



- TCP changes need OS upgrades across the world; QUIC ships inside Chrome/servers — iteration in weeks.
- Encryption is mandatory and integrated: almost the entire packet (incl. handshake) is encrypted.
- Middleboxes see only UDP — they can't ossify or mangle what they can't parse.

STEP 2/5

0-RTT Resumption

Come back to a server you've seen — send data with the FIRST packet.

First visit: 1-RTT handshake

crypto + transport negotiated together — TCP+TLS needed 3+

Return visit: 0-RTT

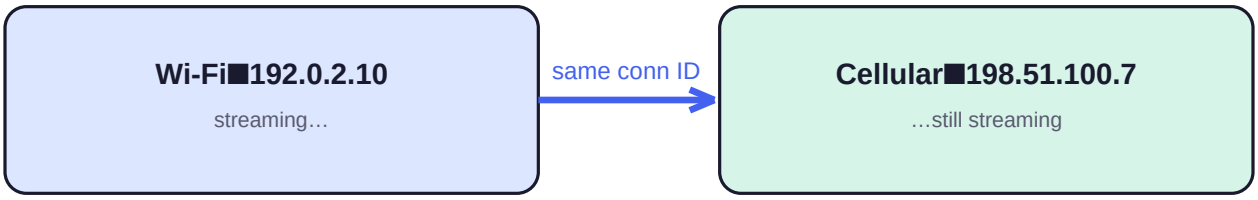
cached keys → application data in the very first UDP datagram

- TCP+TLS 1.3 needs ~2-3 RTTs before data; QUIC does it in 1, then 0 on repeat visits.
- Replay caution: 0-RTT data can be replayed by attackers — servers must gate non-idempotent requests.
- Matters most on high-latency links: each saved RTT is hundreds of milliseconds on mobile.

STEP 3/5

Connection Migration

Your connection has an ID — not an IP:port pair.

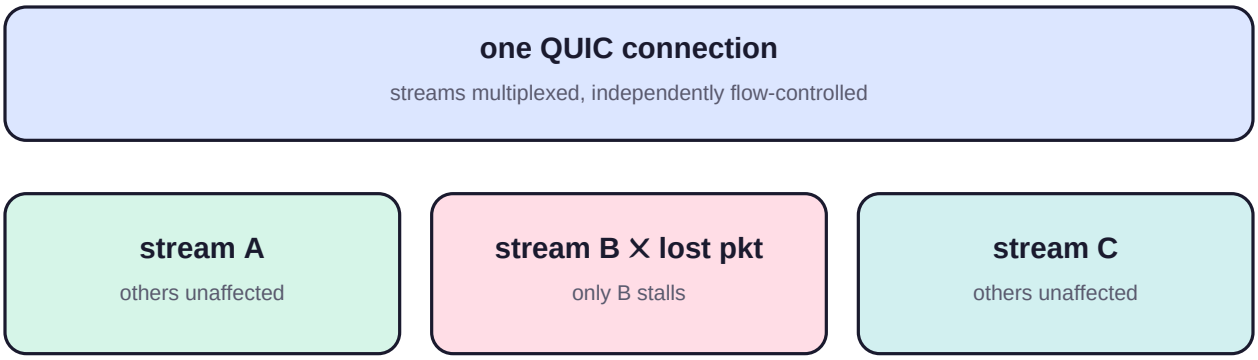


- TCP dies when your IP changes (Wi-Fi → cellular) — the 4-tuple IS the connection.
- QUIC's connection ID survives IP/port changes — the session migrates seamlessly.
- NAT rebinding? Same story: the ID persists while the mapping shifts underneath.

STEP 4/5

Streams Without Head-of-Line Blocking

Independent streams over one connection — loss affects only its stream.

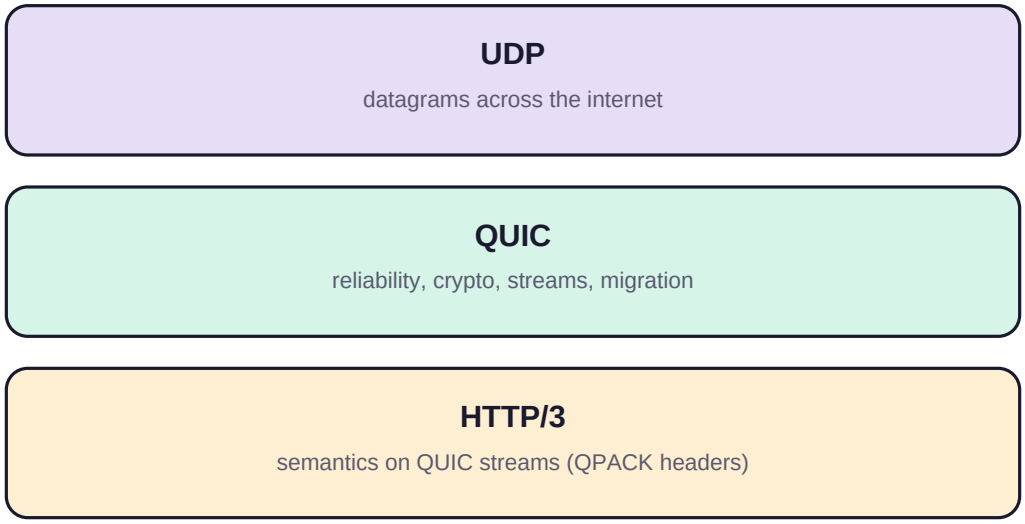


HTTP/2 over TCP: one lost packet froze EVERY stream. QUIC: only the unlucky stream waits.

STEP 5/5

QUIC Is HTTP/3's Foundation

The protocol stack, bottom to top.



QUIC is transport-agnostic in theory — in practice it's the engine under HTTP/3, DNS-over-QUIC, and more.

STEP ✓/✓

Key Takeaways

QUIC — Remember

- QUIC = TCP-like reliability + TLS 1.3, in user space over UDP.
- 0-RTT resumption: repeat visits send data in the first packet (mind replay).
- Connection IDs survive IP changes — Wi-Fi→cellular migration without drops.
- Per-stream loss independence kills TCP's head-of-line blocking.
- QUIC is the transport foundation HTTP/3 is built on.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev